

# Athreya Chandramouli

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## EXPERIENCE

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### Google

Sunnyvale, California, USA

- *Senior Software Engineer (L5) — Data Indexing*

Apr 26 - Present

*Software Engineer III (L4) — Data Indexing*

Nov 24 - Apr 26

- End-to-end project lead for isolating revenue-critical **Ads indexers** into dedicated production stacks; architected parts of the **control plane** (resource estimation, turnup automation, config generation) across hundreds of heterogeneous microservices via an **extensible plugin architecture** over a single source of truth; migrated Ads with **zero data loss and zero downtime, preventing multi-million-dollar outages**.
- Designed and shipped a generic **automatic rollback framework** for a complex microservice indexing stack serving Search, Ads, and YouTube; bypasses manual fault localization, **reducing engineer effort during incidents from hours to ~1 minute** and **cutting total recovery time by 50%**.
- Authored a preemptive proposal to turn up a dedicated isolated stack for **AI training-data indexing workloads** ahead of anticipated demand; quantified blast-radius risk, aligned platform and application team leads across orgs, and **onboarded 6 such applications** onto the new stack, saving weeks of migration time each.

### Google

Dublin, Ireland

- *Software Engineer III (L4) — Ads Quality Infra, Smart Alerting*

Oct 23 - Nov 24

*Software Engineer II (L3) — Ads Quality Infra*

Sep 22 - Oct 23

- Shipped multiple optimizations to the **petabyte-scale observability pipeline** behind Adwords Smart Alerting anomaly detection: **~112% cumulative runtime reduction** and **~170% resource-efficiency gain** (replaced cross-worker shuffles with **in-memory aggregation**, **SIMD** over vectors in lieu of protos, traffic-class upgrade on cross-datacenter reads).
- Engineered a **data versioning system** for the petabyte-scale Adwords retrieval pipeline; unblocked automated backfill and safe data rollbacks, **reducing developer time from ~3 days to ~20 minutes**.
- Service owner for the **Adwords Dynamo extraction service** (service state, capacity planning, cross-team coordination); led incident response on exabyte-scale Adwords Extraction systems, preventing multi-million-dollar outages.

### Rubrik

Remote, India

- *Software Engineering Intern — Hyper-V Infrastructure*

May 21 - Aug 21

- Built a **simulator** for petabyte-scale Microsoft Hyper-V servers, mocking system-level WMI and Apache Thrift RPC calls; enabled end-to-end automated testing and **fault injection, reducing test-bed setup time from ~2-3 days to ~5 minutes**.

## EDUCATION

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### International Institute of Information Technology (IIIT-H)

Hyderabad, India

- *B.Tech + M.S. by Research in Computer Science and Engineering*

Aug 18 - Sep 22

- **GPA:** 9.64/10.0 (Major 9.72/10.0). **Program Gold Medal** (highest GPA in program). **MS Thesis:** *Efficient Parallel and Approximation Algorithms for Percolation Centrality* (Prof. Kishore Kothapalli).
- Coursework: Distributed Systems, Parallel Computing, Operating Systems, Computer Networks, Advanced Algorithms.

## PUBLICATIONS

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- **Efficient Parallel Algorithms for Dynamic Percolation Centrality**. ACM International Conference on Parallel Processing (ICPP), 2025.
- **Coded Data Rebalancing for Distributed Data Storage Systems with Cyclic Storage**. IEEE Information Theory Workshop (ITW), 2022.
- **Efficient Parallel Algorithms for Percolation Centrality**. IEEE International Conference on High Performance Computing (HiPC), 2021.

## SKILLS

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- **Languages:** C++, Golang, C; Python, SQL (familiar).
- **Tooling:** Apache Thrift, Protocol Buffers, MapReduce/Beam (Flume), Bazel, Docker, Git, Linux/Unix, OpenMP, MPI.
- **Focus Areas:** Distributed Systems, Platform Engineering, Performance Optimization, Deployment Pipelines, High-Performance Computing, Agentic Coding (Claude Code, subagent orchestration, custom skills).